Multimodal Prediction of Shunt Responsiveness in Idiopathic Normal Pressure Hydrocephalus

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What is iNPH?

Gait disturbance

Clinical triad

Cognitive impairment

Clinical triad

Urinary dysfunction

Clinical triad

Ventriculomegaly

Imaging feature

Symptoms may improve after CSF diversion.

The complete triad is not required for diagnosis.

- **iNPH matters because it is potentially reversible.**

Why Predict Shunt Responsiveness?

Conditions that mimic iNPH

- Neurodegenerative disorders
- Vascular disease
- Musculoskeletal conditions
- Urologic conditions

Pooled shunt outcomes

74% symptomatic improvement

20.6% complication rate

15.1% revision rate

Salih et al., 2024

■ **Shunting often helps, but patient selection still matters.**

1. Clinical Phenotype

Helpful clues

- Gait-dominant presentation is most informative
- Shorter symptom duration favors improvement
- Baseline function defines meaningful change
- Age alone should not exclude a patient

Main limitation

Symptoms are nonspecific.

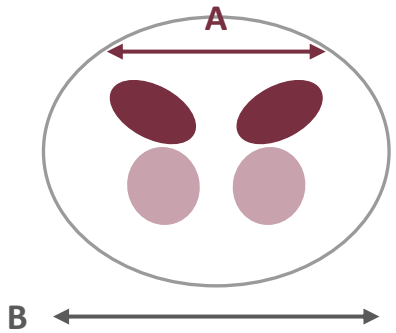
They overlap with many other disorders.

■ **Clinically useful, but not sufficient alone.**

2. Structural Imaging

Evans Index

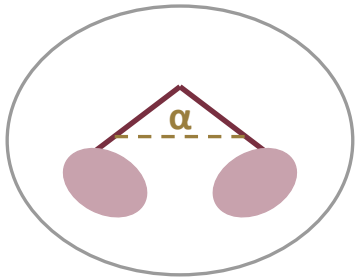
Axial view



- A ÷ B on axial MRI
- > 0.30 supports ventriculomegaly

Callosal Angle

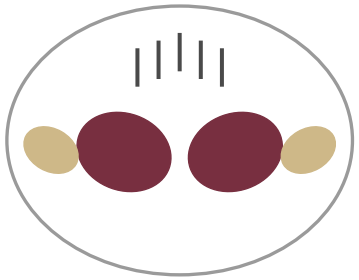
Coronal view



- Measured on coronal MRI
- Smaller angle is more typical of iNPH

DESH

Disproportionately Enlarged Subarachnoid-Space Hydrocephalus



- Enlarged ventricles
- Enlarged Sylvian fissures
- Tight high-convexity sulci

■ **Imaging supports diagnosis better than it predicts shunt response.**

3. Cerebral Spinal Fluid (CSF) Drainage Response

Remove CSF → **Reassess gait or cognition** → **Estimate reversibility**

Tap test	Repeated Lumbar Puncture	External lumbar drainage
Single lumbar CSF removal	Serial testing when equivocal	Continuous drainage over several days

- Positive response strengthens confidence
- Negative response does not reliably exclude benefit

Positive tests help more than negative tests exclude.

4. CSF Hydrodynamics and ICP

ICP = intracranial pressure

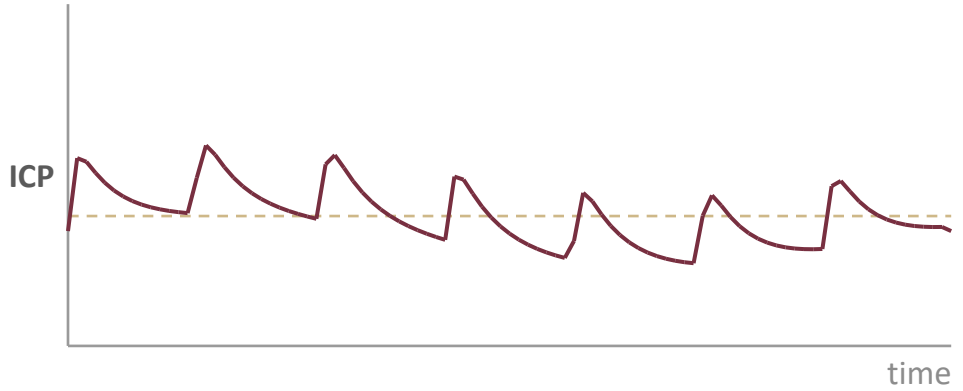
R_{out}
Resistance to CSF outflow

Infusion testing

Controlled infusion used to measure CSF pressure dynamics

ICP monitoring

Continuous assessment of intracranial pressure and pulsatility



Schematic ICP trace — mean level and pulse amplitude

Strength More direct physiologic information

Limitation More invasive, less standardized, less widely available

- **Potentially stronger prediction, but greater complexity.**

5. Biomarkers

Total tau	Higher levels may be associated with non-response
Phosphorylated tau	Higher levels may indicate neurodegenerative burden
Amyloid beta	May reflect overlapping Alzheimer-type pathology, but findings are inconsistent

Promise

Adds biologic context

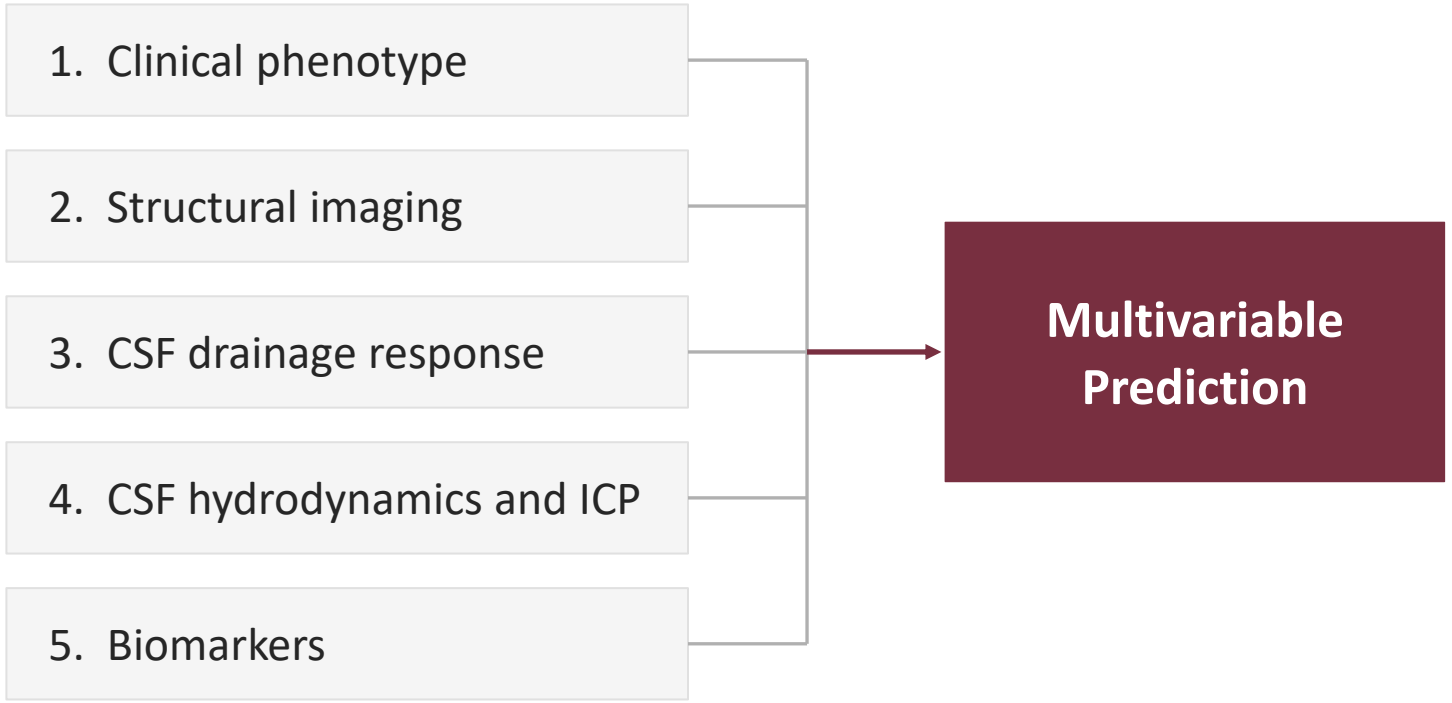
Limitation

No biomarker independently determines candidacy

- **Promising adjuncts, not stand-alone decision tools.**

Next: Conclusion — Multimodal selection

Conclusion: Toward Multimodal Patient Selection



- No single predictor is sufficient
- Each domain provides complementary information
- Future models require standardized prospective validation

Patient selection in iNPH is a multimodal prediction problem.

Thank you — Questions?

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